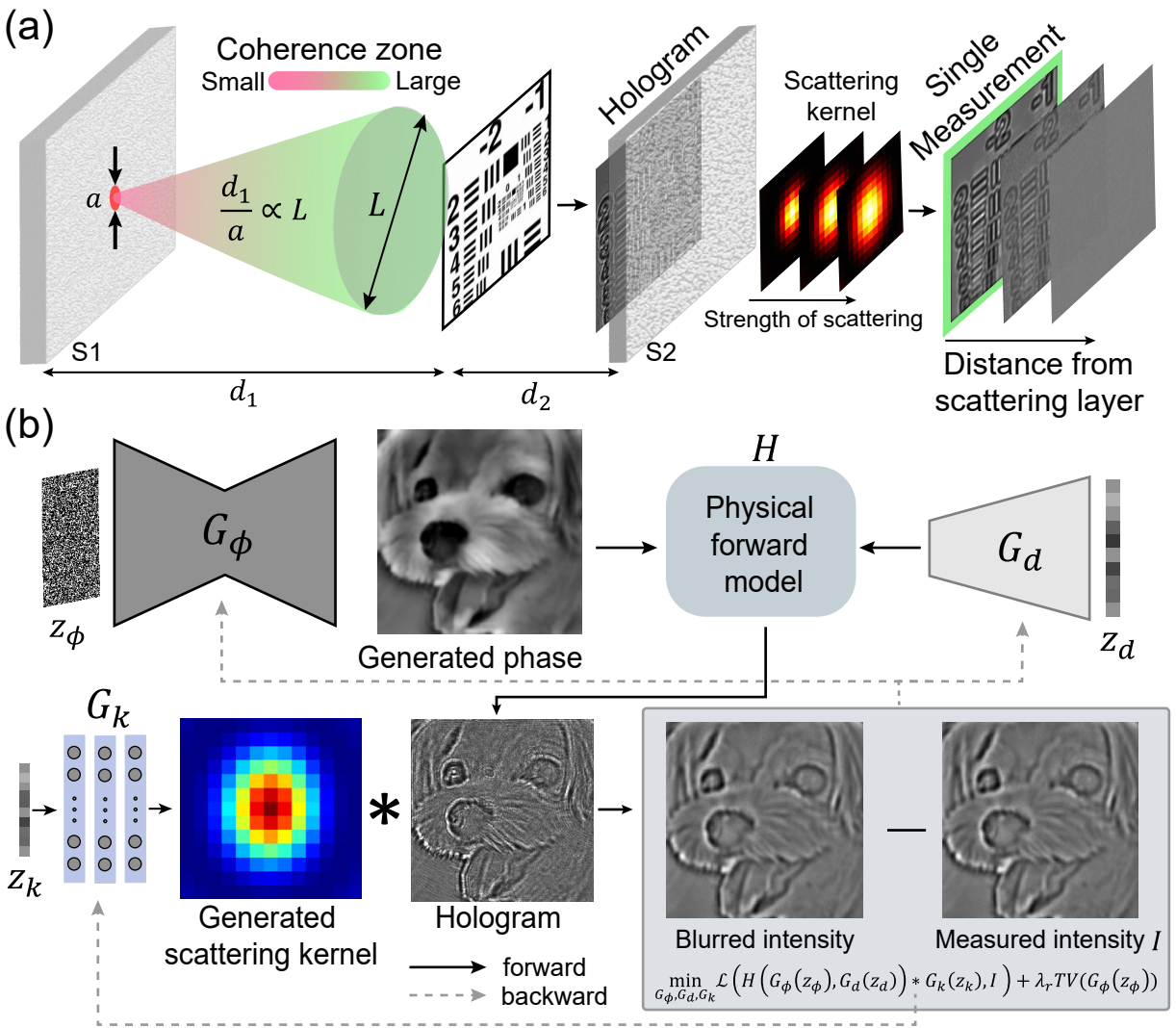
(a)
Coherence zone
Small
Large
a
$\frac{d_1}{a} \propto L$
L
Hologram
Scattering kernel
Single Measurement
Strength of scattering
Distance from scattering layer
S1
S2
d_1
d_2
(b)
G_ϕ
z_ϕ
Generated phase
H
Physical forward model
G_d
z_d
G_k
z_k
Generated scattering kernel
Hologram
Blurred intensity
Measured intensity I
forward
backward
$\min_{G_\phi, G_d, G_k} \mathcal{L}\left(H\left(G_\phi(z_\phi), G_d(z_d)\right) * G_k(z_k), I\right) + \lambda_r TV(G_\phi(z_\phi))$